

Michael Ly, MPH

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SUMMARY

Data engineer building Snowflake Cortex AI and Streamlit tools for technical teams, non-technical users, and legacy systems. AI has moved my bottleneck from writing code to framing problems, breaking them into modular pieces, and designing around how people actually use the tool.

PROFESSIONAL EXPERIENCE

Data Engineer — Ludi Inc | Nashville, TN

Nov 2025 - Present

- Build self-serve internal data tooling in Streamlit in Snowflake, including AI-assisted SQL generation with Snowflake Cortex AI behind a mandatory human-review gate and deterministic validation rules.
- Use Cortex AI to sample columns, draft SQL cleaning logic, and show before-and-after results, with pre and post delta checks to catch data loss.
- Design event-driven, idempotent ingestion on AWS (S3, EventBridge, Lambda, Glue) into Snowflake, with deduplication so failures surface instead of going silent.
- Automate Snowflake schema migrations with Schemachange and DDL diffing that preserves dependent view logic, so changes do not disrupt downstream Power BI reports.
- Build guardrailed production-data tooling with backups before deletes, confirmation gates, audit logging, and reversal scripts, and work on secure data infrastructure for a compliance-sensitive environment.

Data Engineer II — Ellevation Education | Boston, MA

Oct 2024 – May 2025

- Led optimization and stabilization of production Airflow DAGs — reducing pipeline failure rate by 15% through improved error handling, retry logic, and on-call incident response for executive-facing dashboards driving 10% ARR growth.
- Architected and scaled AWS infrastructure (S3, EC2, SFTP, Redshift), reducing ETL runtimes and enabling scalable ingestion for high-volume partner data, saving 20% cloud utilization costs.
- Leveraging CI/CD workflows using Git to automate testing and deployment for reliable, production-ready data pipelining, leading to increased fault tolerance of infrastructure.

Data Engineer I — Ellevation Education

May 2023 – Oct 2024

- Created custom Python workflows by preparing raw partner data from AWS S3 and staging it in Redshift, ensuring accuracy and consistency for downstream analytics.
- Assisted in building and scheduling Airflow DAGs, gaining hands-on experience with pipeline orchestration and automation.

Data Engineer Intern — Brotman Baty Institute

Feb 2022 – Jan 2023

- Contributed to data engineering for the United States' first COVID-19 cases in Seattle, WA.
- Designed and optimized ETL processes using SQL and Python to aggregate survey and swab data from 50,000+ participants, generating over 2M records for epidemiological research.

TECHNICAL SKILLS

Snowflake AI & Apps: Snowflake Cortex AI, Streamlit in Snowflake, AI-assisted data prep, Human-in-the-loop review gates, RBAC, Audit logging

Languages & Tools: Python, SQL, Git, Pandas, Streamlit, Cortex AI, Data Modeling, ETL, SCD2

Cloud & Data Platforms: AWS (Lambda, S3, Glue, EventBridge, EC2, CloudWatch, Athena), Snowflake, Workato

Workflow & Orchestration: Schemachange, dbt, Apache Airflow, SFTP/S3 pipelines, Idempotent event-driven architecture

PROJECTS

[Incremental Data Pipelines with dbt](#) – Seattle, WA

August 2025 - September 2025

- Built incremental dbt models to process only new/updated records, reducing pipeline runtimes and warehouse costs by ~70%.
- Implemented schema validation tests and custom macros to enforce data integrity, detect schema drift, and ensure analytics-ready outputs.

[Comparing Kimball and Inmon Schema Performance in RedShift](#) – Seattle, WA

June 2025 - July 2025

- Benchmarked star vs. snowflake schema performance in AWS Redshift using real-world data volume.
- Created Python-based ETL pipelines for both schemas with data staged in S3 and loaded via COPY commands.
- Used EXPLAIN and query logs to identify storage and performance trade-offs, finding 37% faster reads with Kimball.

EDUCATION & CERTIFICATIONS

MPH in Epidemiology (Data Analytics) from UAB in Birmingham, AL

Status - Graduated

B.A. in Epidemiology from UW in Seattle, WA

Status - Graduated

AWS Certified Solutions Architect Associate

Status - Certified

AWS Certified Data Engineer – Associate

Status - Pending